



Fig. 3: GSM modem

deliver a secondary output of 12V, 250 mA. The transformer output is rectified by bridge rectifier BR1, filtered by capacitor C1 and regulated by IC 7805 (IC3). Capacitor C2 bypasses ripples from the regulated supply. LED1 acts as the power-'on' indicator. Resistor R1 limits the current through LED1.

The AT89C51 microcontroller is connected to the water-level sensor circuit, relay driver and MAX232. The microcontroller is programmed to take necessary actions. The mobile number used in the GSM modem is included in the code before burning the code into the microcontroller.

The water-level sensor circuit comprises transistor T1 (BC548) as sensor driver and water sensors A and B dipped into the borewell along with the pipe. Sensor A is dipped to the threshold point for pumping and sensor B is dipped below the pipe to the bottom of the borewell.

When water in the borewell fills to the threshold level, it is sensed by sensor A and you get a call on your mobile phone. Now you can turn the motor 'on' by sending the SMS "motor11 on" from your mobile phone to the SIM number in the GSM modem. You can also turn the motor 'off' by sending the SMS "motor11 off".

Sensor A is connected to the base of transistor T1 (BC548). When there is a high voltage at the base, T1 con-

ducts and a low voltage is available at its collector. This low signal is fed to pin 12 (port pin p3.2) of the MCU. Similarly, for a low voltage input at

the base, T1 stops conducting and a high voltage signal is available at its collector. So pin 12 of the MCU gets a high signal input. The high or low voltage signal at pin 12 is monitored and processed by the program in the MCU, and decision to turn the motor 'off' taken when the water level dips below sensor A.

Pin 3 (port pin p1.2) of the MCU is the output pin. It is connected to relay-driver transistor T2 (BC548) and LED2. T2 drives relay RL1, which, in turn, activates the motor. LED2 glows to indicate the motor-'on' status.

When water level in the borewell dips below sensor A, the conducting path between sensors A and B breaks. Hence a signal is received by the microcontroller. The microcontroller turns the running motor 'off' and makes a call to the user's cell phone through a GSM modem

TABLE I
Motor, Mobile and LED Status for Different Water Levels

Borewell	Water level	Motor	Mobile	LED2	Remarks
Full	Above point A	On	Call from modem	On	SMS operation to turn on/off the motor
Empty	Below point A	Off	Call from modem	Off	SMS operation not allowed

TABLE II
Key Features of SIM300 Series

Features	Implementation
Power supply	Single supply voltage of 3.4V-4.5V
Power saving	Typical power consumption in SLEEP mode: 2.5 mA
Frequency bands	► SIM300 tri-band (EGSM 900, DCS 1800, PCS 1900). The band can be set by AT COMMAND, and default band is EGSM 900 and DCS 1800 ► Compliant to GSM Phase 2/2+
SMS	► MT, MO, CB, text and PDU mode ► SMS storage: SIM card ► Supports transmission of SMS alternatively over CSD or GPRS. User can choose the preferred mode
SIM interface	Supported SIM card: 1.8V, 3V
External antenna	Connected via a 50-ohm antenna connector or antenna pad
Two serial interfaces	► Serial port 1: Seven lines on the serial port interface ► Serial port 1 can be used for CSD FAX, GPRS and sending AT command of controlling module ► Serial port 1 can use multiplexing function, but you cannot use serial port 2 at the same time ► Autobauding supports baud rate of 1200 to 115,200 bps ► Serial port 2: Two lines on serial port interface, /TXD and /RXD ► Serial port 2 used only for transmitting AT command